How-To Library

Informatica

# Deploy Data Privacy Management on the AWS Cloud Marketplace (10.5.1)

© Copyright Informatica LLC 2021. Informatica and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

# Abstract

You can deploy Data Privacy Management on the Amazon Web Services (AWS) Marketplace. This deployment reference includes step-by-step instructions for deploying Data Privacy Management on the Amazon Web Services (AWS) Marketplace. It also includes information on prerequisites and how to troubleshoot common issues.

# Supported Versions

- Data Privacy Management 10.5.1

# Table of Contents

Overview . . . 2
Intended audience . . . 3
Specialized knowledge . . . 3
Costs and licenses . . . 3
Architecture . . . 4
Deployment Prerequisites . . . 5
Network prerequisites . . . 5
User permissions . . . 6
AWS resources created by the deployment . . . 7
Deploy Data Privacy Management on the AWS Marketplace . . . 8
Deployment steps . . . 9
Step 1. Prepare an AWS account . . . 9
Step 2. Get a license for Data Privacy Management . . . 10
Step 3. Launch the deployment . . . 10
Troubleshooting . . . 18
Log files . . . 18
Verifying AWS CloudWatch Logs for Failures . . . 18
Verifying AWS CloudWatch Logs for Failures . . . 19
Stack Failure . . . 19
Stack Launch Failure When the Pre-validation Check Parameter is Enabled . . . 19
Stack Launch Failure When Rollback On Failure Is Enabled . . . 19
Frequently asked questions on deployment error messages . . . 20
Additional resources . . . 21

# Overview

You can deploy Data Privacy Management on a new Virtual Private Cloud (VPC) or an existing VPC.

Automated reference deployments use AWS CloudFormation templates to launch, configure, and run the AWS compute, network, storage, and other services required to deploy a specific workload on AWS.

In addition, the deployment reference also provides information about the required prerequisites, user permissions, associated costs and licenses, and troubleshooting information.

## Intended audience

This document is for users who are responsible for deploying Data Privacy Management on the AWS Marketplace.

As a user with administrator privileges to deploy applications on AWS, it is assumed that you are familiar with AWS resources such as Cloud formation, VPC, EC2, S3, RDS, Internet gateway, NAT Gateway, Route table, Security group, and Elastic IP.

Familiarity with the concepts of IP CIDR and public and private IP addresses is recommended.

You must also be familiar with Data Privacy Management.

For Data Privacy Management documentation, see the Informatica Documentation Portal.

## Specialized knowledge

Before you deploy Data Privacy Management, we recommend that you become familiar with the following AWS services:

- Amazon VPC
- Amazon EC2
- Amazon RDS
- Amazon S3
- Elastic IP Addresses

If you are new to AWS, see Getting Started with AWS.

## Costs and licenses

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this deployment.

The AWS CloudFormation template for this deployment includes configuration parameters that you can customize. Some of these settings, such as instance type, will affect the cost of deployment. See the pricing pages for each AWS service you plan to use, for cost estimates.

This deployment requires a license for Data Privacy Management. To sign up for a license, contact Informatica.

The following table lists the instance types that you can choose based on sizing requirements:

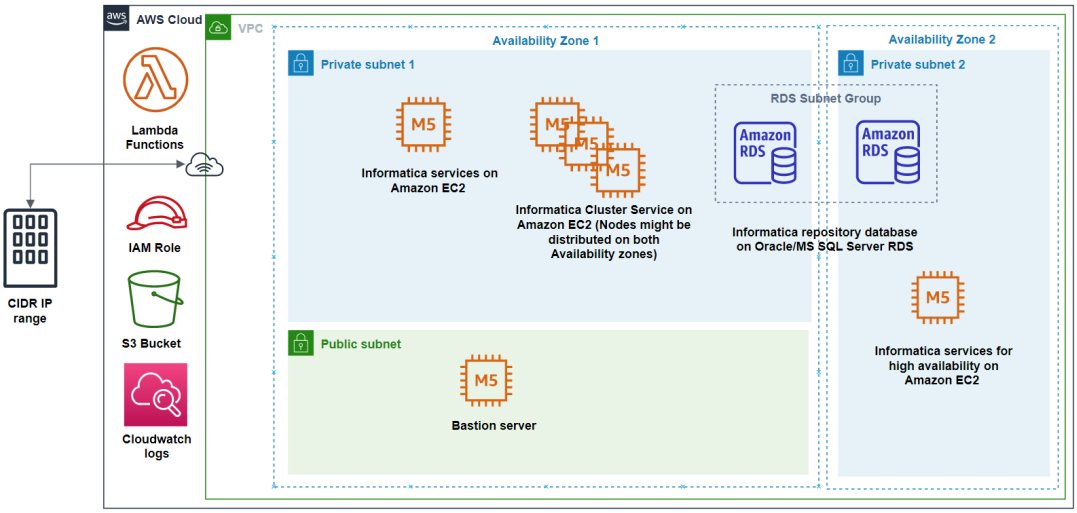
| Virtual Machine | Instance Type | Cluster Size |
|---|---|---|
| Database | M5.xlarge | Small, Medium, Large |
| Informatica Domain | M4.2xlarge / M5.2xlarge | Small, Medium, Large |
| Informatica Cluster Service | M4.4xlarge / M5.4xlarge | Small |
| Informatica Cluster Service | M4.2xlarge / M5.2xlarge | Medium, Large |
| Bastion Server | M5.xlarge/ M4.xlarge | Small, Medium, Large |

The deployment chooses M4 or M5 type of instances based on instance-type availability in a specific AWS region.

# Architecture

When you deploy the Data Privacy Management solution on a new Virtual Private Cloud (VPC) with the default parameters, the deployment builds the following Data Privacy Management environment on the AWS Marketplace:

The following image shows the architecture of Data Privacy Management on AWS Marketplace:




The architecture includes the following components:

- A VPC configured across two Availability Zones. For each Availability Zone, this deployment provisions one private and one public subnet. The Data Privacy Management deployment uses both subnets.
- Managed network address translation (NAT) gateways deployed into the public subnets and configured with an Elastic IP address for outbound internet connectivity.
- An AWS Identity and Access Management (IAM) role with fine-grained permissions for access to AWS services necessary for the deployment process.
- Appropriate security groups for each instance or function to restrict access to only necessary protocols and ports.

In the public subnets, EC2 instances for Data Privacy Management, including the following:

- A single-node or multi-node, embedded cluster based on the cluster size you configure. A node represents an EC2 machine. The Informatica domain runs on a single node with all the associated services. The cluster runs on a separate node or nodes based on your selection. The application services such as the Model Repository Service and the Data Integration Service run on the Informatica domain node.
- Scanners to extract metadata from all data sources supported by Data Privacy Management.
- **Oracle or Microsoft SQL Server database**. You configure the Informatica domain configuration repository, the Model repository, the content management service reference data warehouse, and the profiling warehouse on the database.
- **Informatica domain**. The fundamental administrative unit of the Informatica platform. The Informatica platform has a service-oriented architecture that provides the ability to scale services and share resources across multiple machines.
- **Model Repository Service**. Manages the repository database that stores all the metadata for projects created using Informatica client tools. The Model repository also stores run-time and configuration information for applications that are deployed to a Data Integration Service. The information relating to the administrative module of Data Privacy Management, called Catalog Administrator, is stored in the Model repository.

- **Data Integration Service**. A compute component within the Informatica domain that manages requests submitted to the Hadoop cluster for processing.
- **Informatica Cluster Service**. An application service that runs and manages all the associated services that are required to run Enterprise Data Catalog in the Informatica domain. The associated services include Mongo DB, Nomad, Solr, PostgreSQL, and ZooKeeper.
- **Catalog Service**. Manages Enterprise Data Catalog and manages connections between service components and external applications.
- **Content Management Service**, which manages reference data. It provides reference data information to the Data Integration Service and Informatica Developer.
- **Data Privacy Management Service**. Manages Data Privacy Management and objects in the Data Privacy Management repository.
- **Analyst Service**. Runs the Analyst tool in the Informatica domain. The Analyst Service manages the connections between the service components and the users who log in to the Analyst tool. You can perform column and rule profiling, manage scorecards, and manage bad records and duplicate records in the Analyst tool.
- **Profiling**. Helps you find the content, quality, and structure of data sources of an application, schema, or enterprise. A profile is a repository object that finds and analyzes all data irregularities across data sources in the enterprise and hidden data problems that put data projects at risk. The profiling results include unique values, null values, data domains, and data patterns. In this deployment, profiling can be run on the Data Integration Service (default) or the Informatica cluster.
- **Business Glossary**. Consists of online glossaries of business terms and policies that define important concepts within an organization. Glossaries are stored in a central location for easy lookup by end-users.

You can collect metadata both from resources deployed on a VPC and from resources deployed on premises. For on-premises resources, Data Privacy Management collects metadata through a virtual private network (VPN), a VPC, or by using AWS Direct Connect.

After deploying Data Privacy Management, you must configure the required resources to extract metadata.

If you decide to deploy Data Privacy Management into your existing VPC, the deployment assumes that the infrastructure components already exist, and deploys Data Privacy Management into the environment you specify during deployment. See "Deploy Data Privacy Management on the AWS Marketplace" on page 8 for information on deployments.

# Deployment Prerequisites

Before you deploy Data Privacy Management on the AWS Marketplace, verify that you have performed the following tasks:

- Complete all the network prerequisites.
- Verify the user permissions.

## *Network prerequisites*

Before you deploy Data Privacy Management on the AWS Marketplace, verify that you have completed the network prerequisites for the Data Privacy Management application services:

### VPC and subnet prerequisites

Verify that you have performed the following VPC and subnet prerequisites:

- Set the DNS resolution to **True**.

- Each subnet in the VPC is associated with a route table.
- The routes in the route table must have the route defined to any of the following gateways:
  - Internet Gateway
  - NAT Gateway
  - Virtual Private Gateway
- The subnet associated with Internet Gateway has the Auto-assign public IPv4 property enabled.
- The subnet is associated with any of the following gateways:
  - Internet Gateway
  - NAT Gateway
  - Virtual Private Gateway

### Existing VPC deployment

Verify that you have performed the following prerequisites for an existing VPC deployment:

- The VPC has the DNS resolution enabled.
- Attach the subnet to a route table. Use a local route to the VPC CIDR.
- The service subnets you choose are attached to a route table under the selected VPC.
- The VPC must meet the following requirements:
  - It must be set up with public access through the Internet via an attached Internet gateway.
  - The **DNS Resolution** property of the VPC must be set to **Yes**.
  - The **Edit DNS Hostnames** property of the VPC must be set to **Yes**.

### New VPC deployment

Verify that you have performed the following prerequisites for a new VPC deployment:

- For the availability zone, the selected availability zone must have sufficient capacity to create a new subnet, route table, internet gateway, and NAT gateway.
- The VPC creation must not exceed the supported limit.

## *User permissions*

Before you deploy Data Privacy Management on the AWS Marketplace, verify that you have the required user permissions to launch Data Privacy Management application services on the AWS Marketplace.

Ensure that the AWS user account that you use has the administrator privileges or the following policies assigned to the user account:

- arn:aws:iam::aws:policy/AWSCloudFormationFullAccess
- arn:aws:iam::aws:policy/AmazonEC2FullAccess
- arn:aws:iam::aws:policy/CloudWatchFullAccess
- arn:aws:iam::aws:policy/IAMFullAccess
- arn:aws:iam::aws:policy/AmazonS3FullAccess
- arn:aws:iam::aws:policy/AmazonRDSFullAccess
- arn:aws:iam::aws:policy/AmazonVPCFullAccess
- arn:aws:iam::aws:policy/AWSLambdaFullAccess

# AWS resources created by the deployment

Each deployment creates AWS resources based on the type of deployment.

The following table describes the components that you create when you deploy Data Privacy Management in a new Virtual Private Cloud (VPC):

| Component | Number of Components Created |
|---|---|
| VPC | 1 |
| Internet gateway | 1 |
| Subnet | 2 |
| Route table | 2 |
| Security group | 4 security groups:<br>- 1 for RDS, 1 for Informatica domain instance, and 2 for Informatica Cluster Service nodes<br>- Based on your choice, 1 security group for a Bastion server is created for to support remote log in. The group is created if you selected the option to create the Bastion server as **Yes**. |
| DB subnet group | 1 |
| RDS | 1 Oracle or Microsoft SQL Server RDS instance with 200 GB of storage capacity that runs on a db.m5.xlarge DB instance type. |
| EC2 | The deployment creates the following EC2 components:<br>- One Informatica domain with 1 EC2 instance and 300 GB of storage space. The instance created is of type m5.2xlarge / m4.2xlarge based on availability of the instance size in the selected region.<br>- Based on your choice, one Informatica high availability node with an EC2 instance and 300 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region.<br>- Based on your choice, a small-sized Informatica Cluster with 1 EC2 instance and 300 GB of storage space is created. The instance created is of type m5.4xlarge / m4.4xlarge based on availability of the instance size in the selected region.<br>- Based on your choice, a medium-sized Informatica Cluster with 3 EC2 instances with each instance providing a storage space of 300 GB is created. The instances created are of type m5.2xlarge or m4.2xlarge based on the availability of the instance size in the selected region.<br>- Based on your choice, a large-sized Informatica Cluster with 6 EC2 instances with each instance providing a storage space of 300 GB is created. The instances created are of type m5.2xlarge or m4.2xlarge based on availability of the instance size in the selected region.<br>- Based on your choice, a Bastion server with 1 EC2 instance that has 200 GB of storage space is created. The instance created is of type m5.xlarge or m4.xlarge based on availability of the instance size in the selected region. |
| Elastic IP | 1 Elastic IP attached to the NAT gateway. |
| Internet Gateway | 1 |
| IAM Role and Instance Profile | 1 |

The following table describes the components that you create when you deploy Data Privacy Management in an existing VPC:

| Component | Number of Components Created |
|---|---|
| Security group | 4 security groups:<br>- 1 for RDS, 1 for Informatica domain instance, and 2 for Informatica Cluster Service nodes<br>- Based on your choice, 1 security group for a Bastion server is created for to support remote log in. The group is created if you selected the option to create the Bastion server as **Yes**. |
| DB subnet group | 1 |
| RDS | 1 Oracle or Microsoft SQL Server RDS instance with 200 GB of storage capacity that runs on a db.m5.xlarge DB instance type. |
| EC2 | The deployment creates the following EC2 components:<br>- One Informatica domain with 1 EC2 instance and 300 GB of storage space. The instance created is of type m5.2xlarge / m4.2xlarge, based on availability of the instance size in the selected region.<br>- Based on your choice, one Informatica high availability node with an EC2 instance and 300 GB of storage space. The instance created is of type m5.2xlarge or m4.2xlarge, depending on availability of the instance size in the selected region.<br>- Based on your choice, a small-sized Informatica Cluster with 1 EC2 instance and 300 GB of storage space is created. The instance created is of type m5.4xlarge / m4.4xlarge based on availability of the instance size in the selected region.<br>- Based on your choice, a medium-sized Informatica Cluster with 3 EC2 instances with each instance providing a storage space of 300 GB is created. The instances created are of type m5.2xlarge / m4.2xlarge based on the availability of the instance size in the selected region.<br>- Based on your choice, a large-sized Informatica Cluster with 6 EC2 instances with each instance providing a storage space of 300 GB is created. The instances created are of type m5.2xlarge / m4.2xlarge based on availability of the instance size in the selected region.<br>- Based on your choice, a Bastion server with 1 EC2 instance that has 200 GB of storage space is created. The instance created is of type m5.xlarge / m4.xlarge based on availability of the instance size in the selected region. |
| AWS Lambda Function | 5 AWS Lambda functions to check the feasibility for VPC, subnets, and Informatica license. |
| IAM Role and Instance Profile | 1 |

# Deploy Data Privacy Management on the AWS Marketplace

The automated deployment of Data Privacy Management on AWS Marketplace uses the AWS CloudFormation console to guide your choices and launch the solution deployment.

You can select one of the following deployment options:

- Deploy Data Privacy Management into a new VPC (end-to-end deployment). This option builds and deploys a new AWS environment. The environment that it builds consists of the VPC, subnets, NAT gateways, security groups, and other infrastructure components, and then deploys Data Privacy Management into this new VPC.
- Deploy Data Privacy Management into an existing VPC. This option provisions Data Privacy Management in your existing AWS infrastructure.

The deployment also lets you configure additional settings such as CIDR blocks, instance types, and Data Privacy Management settings, as discussed later in this guide.

Use the AWS CloudFormation console to specify the stack name and to enter parameter values to the template that you use to deploy the application. A stack is a logical representation of an AWS deployment.

Specify the identifiers for the VPC, for the subnet that contains the Amazon EC2 instances where the deployment process deploys the application components, and for the subnet where the deployment process creates the Model repository. The deployment process provisions the application components and creates the Model repository based on the values that you specify.

### Availability Zones and Subnets

The automated solution deploys the components and services for Data Privacy Management into an Amazon VPC configured across two Availability Zones. Each Availability Zone is provisioned with two subnets.

In the first subnet, the deployment process provisions Data Privacy Management application services on an Amazon EC2 instance within the VPC:

- Data Integration Service
- Model Repository Service
- Metadata Access Service
- Informatica Cluster Service
- Content Management Service
- Analyst Service
- Search Service
- Catalog Service
- Data Privacy Management Service

For more information about domain components, see the Informatica Administrator Guide and Informatica Application Service Guide.

In the second subnet, the solution deploys an Oracle database on Amazon RDS for use as the Model repository. The Model repository stores metadata created by Informatica services.

# Deployment steps

You must perform multiple tasks to deploy the Data Privacy Management 10.5.1 AWS Marketplace solution.

You configure multiple parameters, such as availability zones and subnets. The parameters differ depending on whether you deploy the Data Privacy Management on a new VPC or an existing VPC.

## *Step 1. Prepare an AWS account*

Perform the following steps to prepare an AWS account:

1. If you do not have an AWS account, create an account at http://aws.amazon.com by following the on-screen instructions.
2. Use the region selector in the navigation bar to choose the AWS Region where you want to deploy Data Privacy Management on AWS.
3. Create a key pair in your preferred region.
4. When you log in to an EC2 instance, you use a password file for authentication. The file is called a private key file and has a file name extension of .pem.
5. If you do not have an existing .pem key to use, follow the instructions in the AWS documentation to create a key pair.

When you create a key pair, you save the .pem file to your desktop system. Simultaneously, AWS saves the key pair to your account. Make a note of the key pair that you want to use for the Data Privacy Management instance, so that you can provide the key pair name during network configuration.

**Note:** Your administrator might ask you to use an existing key pair.

6. If necessary, request a service limit increase for the EC2 instance type that you want to use for the Informatica domain. You might need to do this if an existing deployment uses this instance type and you think you might exceed the default limit with this reference deployment.

## *Step 2. Get a license for Data Privacy Management*

This deployment requires a license for Data Privacy Management. To sign up for a license, contact Informatica.

1. In your AWS account, create an S3 bucket.
2. Optionally, create a directory under the S3 bucket.
3. Place the Data Privacy Management license key file for the software in the S3 bucket or directory.

## *Step 3. Launch the deployment*

You are responsible for the cost of the AWS services used while running this deployment. There is no additional cost for using this deployment. See the pricing pages for each AWS service you will be using for full details. Each deployment takes approximately one to two hours to complete.

Perform the following steps to launch the deployment:

1. Log in to AWS Marketplace using the AWS account.
2. Type Data Privacy Management in the search box and press **Enter**.
3. Click **Data Privacy Management**. The **Product Overview** page appears.
4. Click **Continue to Subscribe**. The **Subscribe to this software** page appears.
5. Click **Continue to Configuration**. The **Configure this software** page appears.
6. Based on your requirement, select one of the following options from the **Delivery Method** drop-down box to specify where you want to deploy Data Privacy Management:
   - Data Privacy Management (New VPC). Select this option to deploy Data Privacy Management on a new VPC.
   - Data Privacy Management (Existing VPC). Select this option to deploy Data Privacy Management on an existing VPC.
7. Choose **10.5.1** from the **Software Version** list.
8. Click **Continue to Launch**. The **Launch this software** page appears.
9. Select **Launch CloudFormation** from the **Choose Action** drop-down list and click **Launch**. The **Create stack** page appears.
10. Check the region for the deployment displayed in the upper-right corner of the navigation bar, and change it if necessary. This is where the network infrastructure for Data Privacy Management will be built. The template is launched in the U.S. West (Oregon) Region by default.
11. On the **Specify template** section, keep the default setting for the template URL, and then choose **Next**.

    The default option is **Amazon S3 URL**.
12. On the **Specify stack details** page, change the stack name if needed. Review the parameters for the template. Provide values for the parameters that require input.
13. Under **Capabilities**, select the check box to acknowledge that the template will create IAM resources.

14. To deploy the stack, click **Create**.

    Amazon AWS begins to create the stack and displays the CloudFormation dashboard. When the status is CREATE_COMPLETE, the deployment is complete. You can use the URL displayed on the **Output tab** to view the stack properties.

15. For all other parameters, review the default settings and customize them as necessary. When you finish reviewing and customizing the parameters, choose **Next.**

    In the following tables, parameters are listed by category and described separately for the deployment options:

    - "Option 1. Deploy into a new VPC" on page 11
    - "Option 2. Deploy into an existing VPC" on page 13

    **Note:** If you are deploying Data Privacy Management into an existing VPC, make sure that your VPC has a minimum of two subnets. If you choose a remote desktop, you can either use one of the two subnets or use a separate public subnet. These subnets require NAT gateways in their route tables, to allow the instances to download packages and software without exposing them to the Internet. You also need the domain name option configured in the DHCP options, as explained in the Amazon VPC documentation. When you launch the deployment, the wizard prompts you for the VPC settings.

## Option 1. Deploy into a new VPC

Specify the identifiers for the VPC, for the subnet containing the Amazon EC2 instances where the deployment process deploys the application components, and for the subnets used by the Informatica services. The deployment process provisions the application components and creates the required repository databases based on the values you specify.

1. Enter the name of the stack to create. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).
2. Enter the parameters required to deploy the application.
   The following table lists the parameters in the **Network Configuration** section:

| Parameter | Default | Description |
|---|---|---|
| Availability Zones | Requires input | The list of Availability Zones to use for the subnets in the VPC. Make sure that you select two zones. |
| VPC CIDR | 10.0.0.0/16 | Classless Inter-Domain Routing IP address range associated with the Amazon VPC.<br>The CIDR range must have permissions to access the public subnet where the application runs.<br>Format the string as `<IP address>/<IP network prefix>`. For example, to specify a range from `10.20.30.40` to `10.20.30.47`, enter `10.20.30.40/29`. |
| Public Subnet CIDR | Requires input | CIDR block for the public subnet. Ensure that you include the subnet in the VPC CIDR. |
| Private Subnet 1 CIDR | 10.0.0.0/19 | CIDR block for the private subnet 1. Ensure that you include the subnet in the VPC CIDR. |

| Parameter | Default | Description |
|---|---|---|
| Private Subnet 2 CIDR | 10.0.32.0/19 | CIDR block for the private subnet 2. Ensure that you include the subnet in the VPC CIDR. |
| CIDR IP Range | Requires input | The CIDR IP range that is permitted to access Data Privacy Management.<br>To reduce the potential of inbound attacks from unknown IP addresses, you can use a constrained CIDR range to reduce. For example, to specify the range of 10.20.30.40 to 10.20.30.47, enter 10.20.30.40/29. |

The following table lists the parameters in the **Bastion Server for Informatica** section:

| Parameter | Default | Description |
|---|---|---|
| Deploy Bastion Server | Requires input | Select **Yes** to deploy a bastion server to access other resources in the VPC. |

The following table lists the parameters in the **Amazon EC2 Configuration** section:

| Parameter label (name) | Default | Description |
|---|---|---|
| Key Pair Name | Requires input | Name of an existing Amazon EC2 key pair to enable SSH and Remote Desktop Protocol (RDP) access to Data Privacy Management. |
| Keypair S3 URI | Requires input | S3 Bucket URI for the keypair selected. Use the following format:<br>`s3://bucket-name/Keypair.pem` or<br>`s3://bucket-name/sub-folder/Keypair.pem` |

The following table lists the parameters in the Informatica Data Privacy Management section:

| Parameter label (name) | Default | Description |
|---|---|---|
| License Key S3 Bucket URI | Requires input | The S3 bucket URI for the Data Privacy Management license key in the following format: `s3://<bucket-name>/<directory>/<license key>` |
| Database Type | Oracle | Select the database type for the Informatica repository and services. Choose Oracle or Microsoft SQL Server. |
| High Availability Deployment | Requires input | Enable or disable high availability for the deployment.<br>For information about high availability for the Informatica domain, see the Informatica documentation. |

| Parameter label (name) | Default | Description |
|---|---|---|
| Deployment Type | Requires input | Choose from the following options for the Informatica cluster:<br>- Small: one m*.4xlarge node<br>- Medium: three m*.2xlarge nodes<br>- Large: six m*.2xlarge nodes<br>**Note:** The instance types you can select might depend on your AWS region. |
| Use S3 files | No | Use configuration files stored on an Informatica-owned S3 bucket. Selecting Yes downloads configuration files and then configures. May contain latest fixes if any, else the deployment uses local configuration files stored in the EC2 instance. |

3. Click **Next**.
4. On the Options page, expand the **Advanced** tab.
5. Set the **Rollback on Failure** property to **No**.
6. Click **Next**.
7. Click **Create**.
8. When the status for the stack reads CREATE_COMPLETE, click the stack name.
9. Click the **Outputs** tab to find the URLs for the Informatica Administrator, Catalog, and Data Privacy Management.

## Option 2. Deploy into an existing VPC

1. Enter a name for the stack. The stack name can include letters (A-Z and a-z), numbers (0-9), and dashes (-).
2. Enter the parameters required to deploy the application.
   The following table lists the parameters in the **Network Configuration** section:

| Parameter | Default | Description |
|---|---|---|
| VPC | Requires input | ID of your existing VPC where you want to deploy Data Privacy Management. |
| Subnet 1 | Requires input | Select the subnet ID where you want to deploy Data Privacy Management. Make sure that the subnet is included in the VPC and associated with an Internet Gateway or a NAT Gateway. Ensure that service subnet ID1 and service subnet are not the same. |
| Subnet 2 | Requires input | If you want to deploy Data Privacy Management with high availability enabled, provide the subnet ID that you want to use for high availability. Make sure that subnet 1 and subnet 2 are in different availability zones. |

| Parameter | Default | Description |
|---|---|---|
| CIDR IP Range | Requires input | The CIDR IP range that is permitted to access Data Privacy Management. To reduce the potential of inbound attacks from unknown IP addresses, you can use a constrained CIDR range. For example, to specify the range of 10.20.30.40 to 10.20.30.47, enter 10.20.30.40/29. |
| Pre-validation Check | Requires input for a | Select to validate the VPC and subnets before deploying Data Privacy Management.<br>If you choose to have the automated solution perform a validation check before launching the deployment, the check validates your parameter inputs. If the deployment as specified fails the check, the deployment process stops. The deployment logs record the problem. |

The following table lists the parameters in the **Bastion Server for Informatica** section:

| Parameter | Default | Description |
|---|---|---|
| Deploy Bastion Server | Requires input | Select **Yes** to deploy a bastion server to access other resources in the VPC. |
| Bastion Subnet | Requires input | Select the subnet for the bastion server if you selected **Yes** for the previous option. Ensure that the subnet is included in the VPC that you use and attached to an Internet Gateway. |

The following table lists the parameters in the **Amazon EC2 Configuration** section:

| Parameter label (name) | Default | Description |
|---|---|---|
| Key Pair Name | Requires input | Name of an existing Amazon EC2 key pair to enable SSH and Remote Desktop Protocol (RDP) access to Data Privacy Management. |
| Keypair S3 URI | Requires input | S3 Bucket URI for the keypair selected. Use the following format:<br>`s3://bucket-name/Keypair.pem` or<br>`s3://bucket-name/sub-folder/Keypair.pem` |

The following table lists the parameters in the Informatica Data Privacy Management Configuration section:

| Parameter label (name) | Default | Description |
|---|---|---|
| License Key S3 Bucket URI | Requires input | The S3 bucket URI for the Data Privacy Management license key in the following format: `s3://<bucket-name>/<directory>/<license key>` |
| Database Type | Oracle | Select the database type for the Informatica repository and services. Choose Oracle or Microsoft SQL Server. |
| High Availability Deployment | Requires input | Enable or disable high availability for the deployment.<br>For information about high availability for the Informatica domain, see the Informatica documentation. |

| Parameter label (name) | Default | Description |
|---|---|---|
| Deployment Type | Requires input | You can select one of the following sizes:<br>- Select the Amazon EC2 instance type where you deploy the Informatica cluster.<br>Choose from the following options for the Informatica cluster:<br>- Small: one m*.4xlarge node<br>- Medium: three m*.2xlarge nodes<br>- Large: six m*.2xlarge nodes<br>**Note:** The instance types you can select might depend on your AWS region. |
| Use S3 files | No | Use configuration files stored on an Informatica-owned S3 bucket. Selecting Yes downloads configuration files and then configures. May contain latest fixes if any, else the deployment uses local configuration files stored in EC2 instance. |

3. Click **Next**.
4. On the **Configure stack options** page, you can specify tags (key-value pairs) for resources in your stack and set advanced options. When you are done, choose **Next**.
5. On the **Review** page, review and confirm the template settings. Under Capabilities, select the check box to acknowledge that the template will create IAM resources.
6. Choose **Create** to deploy the stack.
7. Monitor the status of the stack. When the status is `CREATE_COMPLETE`, the deployment is complete.

You can use the keys displayed in the **Output** page as shown in the following table to view the details of the deployment:

| Key | Description |
|---|---|
| AdminConsoleUser | Username to access the Informatica domain. |
| AdminConsoleUserPwd | Password for the Informatica domain username. |
| BastionPrivateDNS | Private DNS of the Informatica bastion server. |
| BastionPublicIPDNS | Public DNS of the Informatica bastion server. |
| DBUserPwd | Password to access the Informatica domain repository database. |
| DBUsers | Usernames to access the Informatica domain repository database. |
| HANodeIP | IP address of the Informatica high availability node. |
| InformaticaAdminConsolePrivateURL | Private URL to access the Informatica Administrator. |
| InformaticaAdminConsolePublicURL | Public URL to access the Informatica Administrator. |
| informaticaAdminConsoleServerLogs | Location where the Informatica domain installation log files are stored. |
| InstanceSetupLogs | Location where the Informatica domain instance setup log files are stored. |
| NodePrivateEDCAdminURL | Private URL to access Catalog Administrator. |

| Key | Description |
|---|---|
| NodePrivateEDCCatalogURL | Private URL to access Enterprise Data Catalog. |
| NodePrivateDPMURL | Private URL to access Data Privacy Management. |
| NodePublicEDCAdminURL | Public URL to access Catalog Administrator. |
| NodePublicEDCCatalogURL | Public URL to access Enterprise Data Catalog. |
| NodePublicInfaDPMURL | Public URL to access Data Privacy Management. |
| AdvScannerPrivateURL | Private URL to access Informatica EDC Advanced Scanner |
| AdvScannerPublicURL | Public URL to access Informatica EDC Advanced Scanner |
| AdvScannerUser | Username to access Informatica EDC Advanced Scanner |
| AdvScannerUserPwd | Password for the Informatica EDC Advanced Scanner user name |

See the Data Privacy Management User Guide for information about logging in to Data Privacy Management.

The following table shows a sample completed stack for Data Privacy Management deployed on AWS:

| Stack and Nested Stack Names | Stack Description | Condition for Stack Usage |
|---|---|---|
| StackName | Main stack | Not applicable |
| StackName-LicenseCheckLambdaStack | Template to verify if the license key file is valid. | Default |
| StackName-KeypairCheckLambdaStack | Checks whether the keypair file is valid. | Created by default.<br>**Note:** Applicable only on Linux deployments |
| StackName-BastionSubnetCheckLambdaStack | Template to verify the validity of a subnet. | Default. Creates the AWS Lambda function if the **Pre-validation Check** and the **Bastion Server** parameters are set to **Yes**. |
| StackName-SubnetsAZCheckLambdaStack | Template to verify if the subnets are in different availability zones. | Default |
| StackName-PublicIpDnsCheckLambdaStack | Template to verify if an auto-assign public IPv4 address is enabled for subnet. | Default |
| StackName-Subnet2CheckLambdaStack | Template to verify the validity of a subnet. | Default. Creates the AWS Lambda function if the **Pre-validation Check** parameter is set to **Yes**. |
| StackName-Subnet1CheckLambdaStack | Template to verify the validity of a subnet. | Default. Creates the AWS Lambda function if the **Pre-validation Check** parameter is set to **Yes**. |

| Stack and Nested Stack Names | Stack Description | Condition for Stack Usage |
|---|---|---|
| StackName-IAMStack | Template to create the IAM roles and profiles used by EC2 instances. | Default |
| StackName-SecurityGroupStack | Template to create the security groups and network interfaces used by EC2 instances. | Default |
| StackName-RDSStack | Template to create the RDS resources. | Default |
| StackName-BastionServerStack | Template to create the bastion server EC2 instances. | Used when the **Deploy Bastion Server** parameter is set to **Yes**. |
| StackName-ICSStack | Template to create the Informatica cluster. | Default |
| StackName-ICSStack-ICSGatewayStack | Template to create the Informatica cluster gateway node. | Default |
| StackName-ICSStack-Infa-ICS-Node2 | Template to create the Informatica cluster node. | Used when the **Deployment Type** parameter is set to **Medium**. |
| StackName-ICSStack-Infa-ICS-Node3 | Template to create the Informatica cluster node. | Used when the **Deployment Type** parameter is set to **Medium**. |
| StackName-ICSStack-Infa-ICS-Node4 | Template to create the Informatica cluster node. | Used when the **Deployment Type** parameter is set to **Large**. |
| StackName-ICSStack-Infa-ICS-Node5 | Template to create the Informatica cluster node. | Used when the **Deployment Type** parameter is set to **Large**. |
| StackName-ICSStack-Infa-ICS-Node6 | Template to create the Informatica cluster node. | Used when the **Deployment Type** parameter is set to **Large**. |
| StackName-DPMServerStack | Template to create the Informatica Data Privacy Management node. | Default |
| StackName-DPMHANodeStack | Template to create the Informatica Data Privacy Management high availability node. | Used when the **High Availability Deployment** option is enabled. |

## Accessing Informatica Administrator

Based on the type of deployment, follow the instructions to access Informatica Administrator:

- Deployment on a new VPC. Use an instance in the same VPC that you can access from outside the VPC. It is assumed that you have not deployed a remote Windows server to access other resources in the VPC. Use that instance to access other resources. See the following URL formats that you can use:
  - To access the Informatica Administrator, use the URL in the following format: <Private IP Address or private DNS>:6005
  - To access Data Privacy Management use the URL in the following format: *<Private IP Address or private DNS>:18295*

- Deployment on an existing VPC. If the created instances are assigned a public IP address or DNS, you can access the Informatica Administrator using the public IP address or the DNS with port numbers as shown in the following sample:
  - To access the Informatica Administrator, use the URL in the following format: *<Public IP Address or DNS>:8443*
  - If the public IP address or DNS is not assigned to the created instance, then you can use the private IP address or DNS. Use an instance in the same VPC if you have not deployed a remote Windows server to access other resources in the VPC. Make sure that you can access the instance from outside the VPC. Use that instance to access other resources. For example, to access the Informatica Administrator , use the URL in the following format:*<Private IP Address or private DNS>:6008*.
  - To access Data Privacy Management use the URL in the following format: *<Private IP Address or private DNS>:18295*

# Troubleshooting

You can review the AWS log files to troubleshoot issues with your Data Privacy Management deployment. Use the CloudWatch service to monitor all of the events in your AWS deployments.

## *Log files*

Log files for troubleshooting are available in the following locations in the Informatica server:

- Node preparation log file: `/opt/Infa_node_configure.log`
- Installation log file: `/opt/Infa_OneClick_Solution.log`
- Service Initialization log file: /var/log/cfn-init.log
- Informatica Cluster Service log file: /<opt/Informatica/<Infa_Oracle or Infa_MSSQLServer based on database selected>/logs/InfaNode/services/InfaHadoopService/ICS/ICS.log
- Catalog Service log file: /opt/Informatica/<Infa_Oracle or Infa_MSSQLServer based on database selected>/logs/InfaNode/services/CatalogService/Catalog_ Service/LDM.log
- Data Privacy Management log file: /opt/Informatica/<Infa_Oracle or Infa_MSSQLServer based on database selected>/logs/InfaNode/services/SecureAtSourceService/Data_Privacy_Management_Service/SecureAtSourceService.log

If the Informatica EC2 instance is running, but the stack fails or if the Informatica Administrator console does not open, see the following log files to identify the issues:

- /opt/Infa_OneClick_Solution.log
- /var/log/cfn-init-cmd.log
- /var/log/cfn-init.log
- /var/log/cloud-init.log

## *Verifying AWS CloudWatch Logs for Failures*

Perform the following steps to verify the AWS CloudWatch logs if any of the Lambda stacks fail:

1. In the Amazon Web Services Console, select **CloudWatch** from the **Services** menu.
2. In the navigation pane, choose **Logs** > **Log Groups**.
3. Enter `/aws/lambda/<stack name>` in the search pane.
4. Select the log group associated with the name of the stack that failed.

## Verifying AWS CloudWatch Logs for Failures

CloudWatch is an AWS service where you can monitor all of the events in your AWS deployments. CloudWatch provides a central location from which to monitor EC2 instances and other resources. For more information, see the AWS documentation.
Perform the following steps to access the Data Privacy Management deployment log files using the AWS console:

1. In the Amazon Web Services Console, select **CloudWatch** from the **Services** menu.
2. In the navigation pane, select **Logs** > **Log Groups**.
3. Search for Data Privacy Management in the search pane.

### AWS CloudWatch Log Files for Specific Stack Failure Scenarios

See the following AWS CloudWatch log files for the following specific stack failure scenarios:

- Stack fails when the **Pre-validation Check** parameter is set to **Yes**. Search for `/aws/lambda/<Stack-Name>` in the **Log Groups** search box and then click the log group to view the log files for the stack.
- Stack fails when the **Rollback on failure** option is enabled by default. Search for `InformaticaDPMLogGroup` in the **Log Groups** search box and click any of the following log groups:
  - installation.log
  - cfn-init
  - InformaticaClusterService
  - Catalog Service

## Stack Failure

Review the following log files if an EC2 instance where Informatica services run is running but the stack fails, or if the Administrator tool fails to launch:

- Node preparation log file: `/opt/Infa_node_configure.log`
- Installation log file: `/home/Infa_OnceClick_Solution.log`
- Command Initialization log file: `/var/log/cfn-init-cmd.log`
- Service Initialization log file: `/var/log/cfn-init.log`
- Cloud log file: `/var/log/cloud-init.log`

## Stack Launch Failure When the Pre-validation Check Parameter is Enabled

Review the AWS CloudWatch log files if a stack failure occurs when the Pre-validation Check option is enabled.

1. In the AWS Management Console, select **CloudWatch** from the **Services** menu.
2. Select **Logs**, then and then select **Log Groups**.
3. Enter the following string in the search bar:

   `/aws/lambda/<stack name>`
4. Click the log group associated with the stack.

## Stack Launch Failure When Rollback On Failure Is Enabled

Review the AWS CloudWatch log files if a stack failure occurs when the **Rollback on Failure** option is enabled.

1. In the AWS Management Console, select **CloudWatch** from the **Services** menu.
2. Select **Logs**, then and then select **Log Groups**.

3. Enter the following string in the search bar: `InformaticaDPMLogGroup`
4. Click a log file in the log group associated with the stack.

   You might want to review the following log files:

   - installation.log
   - cfn-init
   - InformaticaClusterService
   - EDCService
   - DPMService

# Frequently asked questions on deployment error messages

**I encountered a CREATE_FAILED error when I launched the deployment.**

If AWS CloudFormation fails to create the stack, we recommend that you relaunch the template with **Rollback on failure** set to **No**. (This setting is under **Advanced** in the AWS CloudFormation console, **Options** page.) With this setting, the stack's state will be retained, and the instance will be left running, so you can troubleshoot the issue. (You will want to look at the log files in `/var/log/cfn-init-cmd.log`)

**Note:** When you set **Rollback on failure** to **No**, you will continue to incur AWS charges for this stack. Please make sure to delete the stack when you have finished troubleshooting.

For additional information, see Troubleshooting AWS CloudFormation on the AWS website or contact us on the AWS Discussion Forum.

**I encountered errors while installing the Informatica domain and services. Where can I find more information?**

View the `/opt/Infa_OneClick_Solution.log` log file to see more information about the errors that you encountered.

**I encountered a size limitation error when I deployed the AWS CloudFormation templates.**

We recommend that you launch the deployment templates from the location we provided or from another S3 bucket. If you deploy the templates from a local copy on your computer or from a non-S3 location, you might encounter template size limitations when you create the stack. For more information about AWS CloudFormation limits, see the AWS documentation.

**What should I do before I relaunch the template with Rollback on failure?**

Ensure that you disable rollback when you launch the stack in case of a stack failure. If Data Privacy Management server stack is created, check the following logs to identify the issues:

- `/opt/Infa_OneClick_Solution.log`
- `/var/log/cfn-init.log`
- `/opt/Informatica/<Infa_Oracle or Infa_MSSQLServer based on database selected>/logs/InfaNode/services/InfaHadoopService/ICS/ICS.log`
- `/opt/Informatica/<Infa_Oracle or Infa_MSSQLServer based on database selected>/logs/InfaNode/services/CatalogService/Catalog_Service/LDM.log`
- `/opt/Informatica/<Infa_Oracle or Infa_MSSQLServer based on database selected>/logs/InfaNode/services/SecureAtSourceService/Data_Privacy_Management_Service/SecureAtSourceService.log`

If the Data Privacy Management server stack is not created, you can check the CloudWatch logs to identify the issues. For more information, see the Verifying AWS CloudWatch Logs for Failures section.

**I am facing Model Repository Service connectivity issues with the following error messages:**

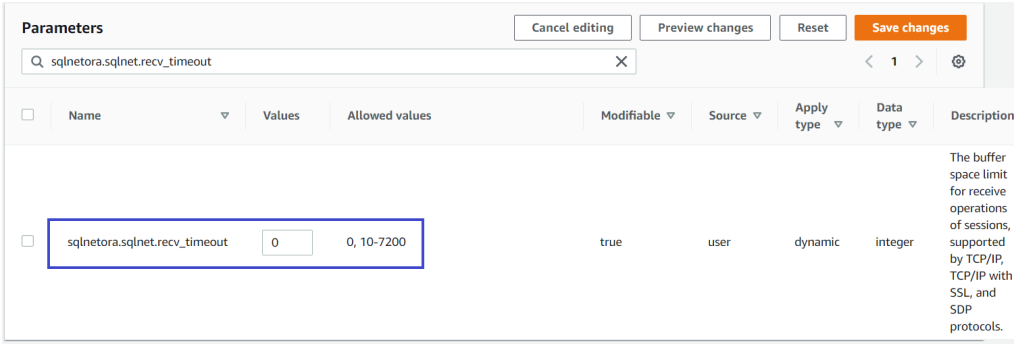
```
MRS_50015 "The Repository Service operation failed. ['[PRSVCSHARED_01707] Connection
issues with the configured database. Request failed with the error message
[[PERSISTENCEAPI_0307] [DBPERSISTER_1005] Failed to process requested operation. This
was caused by [informatica][Oracle JDBC Driver]No more data available to read.].']"
```

```
MRS_50015 "The Repository Service operation failed. ['[PRSVCSHARED_01707] Connection
issues with the configured database. Request failed with the error message
[[PERSISTENCEAPI_0307] [PERSISTENCECOMMON_0001] Internal error. The request processing
failed. This was caused by [informatica][Oracle JDBC Driver]Exception generated during
deferred local transaction handling.  See next exception via
SQLException.getNextException for details.].']"
```

Applicable for Oracle. Perform the following steps to resolve the Model repository database connectivity issue:

1. Launch the AWS RDS page.
2. Select the RDS instance.
3. Click the **Configuration** tab.
4. Scroll down the tab page and click the link under the **Parameter group** section.
5. Search for the following parameters in the list of parameters that appear:
   - sqlnetora.sqlnet.send_timeout
   - sqlnetora.sqlnet.recv_timeout
   - sqlnetora.sqlnet.inbound_connect_timeout
   - sqlnetora.sqlnet.outbound_connect_timeout
   - sqlnetora.tcp.connect_timeout

   **Note:** If the listed parameters are not present in the list, make sure that add the parameters to the list.
6. Change the value of each of the parameters listed in the previous step to 0 as shown in the following sample image:
   The following image shows the `sqlnetora.sqlnet.recv_timeout` parameters




7. Click **Save changes**.

# Additional resources

## AWS Services

- AWS CloudFormation
- Amazon EBS
- Amazon EC2

- Amazon VPC

### Data Privacy Management Documentation

- Product Documentation

### Reference Deployments

- AWS Deployment home page

## Author

**Sadhana Kamath**